

Unit-II Conventional and Symmetric Cryptography

2.1 Introduction:

1. Plain text

- ✓ It is any readable data — including binary files — in a form that can be seen or utilized without the need for a decryption key or decryption device.
- ✓ Data that can be read and understood without any special measure.
- ✓ Plaintext would refer to any message, document, file, and the like intended or having been encrypted.
- ✓ Plaintext is the input to a crypto system, with cipher text being the output.
- ✓ In cryptography, algorithms transform plaintext into cipher text, and cipher text into plaintext.

2. Cipher text

- ✓ Cipher text is encrypted text transformed from plaintext using an encryption algorithm.
- ✓ Cipher text can't be read until it has been converted into plaintext (decrypted) with a key.

3. Cryptography or Cryptology

- ✓ Cryptography is a Greek word having the meaning of “**Secret Writing**”. It is the science of using mathematics to encrypt and decrypt data.
- ✓ **Cryptography** focuses on creating secret codes for providing security to information.

4. Cryptanalysis

- ✓ It is the breaking of “Secret Codes”.
- ✓ It is the science of breaking Encryption.
- ✓ Cryptanalysis which is the study of the cryptographic algorithm and the breaking of those secret codes.

5. Encryption

- ✓ Encryption is the process by which a readable message is converted to an unreadable form to prevent unauthorized user from reading it.
- ✓ It is an Algorithm for transforming plain text to cipher text.

6. Decryption

- ✓ Decryption is the process of converting an encrypted message back to its original (readable) format. The original message is called the plaintext message.
- ✓ It is an Algorithm for transforming cipher text to plain text.

Symmetric key cryptography/Encryption	Asymmetric key cryptography/Encryption
It is also known as Conventional Cryptography/Secret Key /Private Key Cryptography.	It is also known as Public key cryptography.
Both sender and receiver use single same key for Encryption and Decryption.	It uses two keys: 1. Public key known to everyone. 2. Private key known to receiver.
Example: DES(Data Encryption Standard),AES	Example: RSA algorithm, DSA
Key distribution poses serious problem.	Key distribution does not pose serious problem.
The encryption process is very fast.	The encryption process is slow.
Security is less as only one key is used for both encryption and decryption purpose.	It is more secure as two keys are used here- one for encryption and the other for decryption.
Less complexity of algorithm.	More complexity of algorithm.

2.2 Substitution and Transposition Techniques

❖ Substitution Techniques

- ✓ Substitution technique involves replacing letters with other letters and symbols. In simple terms, the characters present in the initial message are restored by the other characters or numbers or by symbols.
- ✓ There are various types of substitution ciphers which are as follows:

1. Caesar Cipher
2. Playfair Cipher
3. Hill Cipher
4. One Time Pad

1. Monoalphabetic Cipher

1. Monoalphabetic Cipher:

A monoalphabetic cipher is any cipher in which the letters of the plain text are mapped to cipher text letters based on a single alphabetic key. Examples of monoalphabetic ciphers would include the **Caesar-shift cipher**, where each letter is shifted based on a numeric key.

2. Polyalphabetic Cipher

A polyalphabetic cipher is any cipher based on substitution, using multiple substitution alphabets. The **Vigenère cipher** is probably the best-known example of a polyalphabetic cipher, though it is a simplified special case.

❖ Caesar Cipher:

- It is also known as **shift** cipher or **additive** cipher.
- It is a type of substitution cipher in which each letter in the plaintext is 'shifted' a certain number of places down the alphabet.
- For example, with a shift of 1, A would be replaced by B, B would become C, and so on.
- First translate all of characters to numbers, 'a'=0, 'b'=1, 'c'=2, ... , 'z'=25.

For Encryption, $C=E(P)=(P+K) \bmod 26$

For Decryption, $P=D(C)=(C-k) \bmod 26$

Example:

Plain Text=HELLO AND Key=3

Encryption:

Cipher Text=KHOOR

Decryption:

Plain Text=HELLO

Example :- ① caesar cipher
Plain Text = "hello" and key = 3

⇒ Encryption :- $E(P) = (P + k) \bmod 26$

a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

P.T. = hello, $k=3$ (key)

$$\begin{aligned} E(h) &= (h + k) \% 26 \\ &= (7 + 3) \% 26 \\ &= 10 \\ &= k \end{aligned}$$

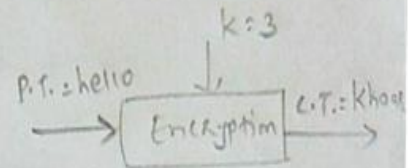
$$\begin{aligned} E(e) &= (e + k) \% 26 \\ &= (4 + 3) \% 26 = 7 = h \end{aligned}$$

$$\begin{aligned} E(l) &= (l + k) \% 26 \\ &= (11 + 3) \% 26 = 14 = o \end{aligned}$$

$$E(o) = 14 \% 26 = 14 = o$$

$$\begin{aligned} E(o) &= (o + k) \% 26 \\ &= (14 + 3) \% 26 = 17 = r \end{aligned}$$

∴ C.T. = khoox



⇒ Decryption :- $D(C) = (C - k) \bmod 26$

C.T. = Khoox, key = 3

$$\therefore D(k) = (k - 3) \% 26 = (10 - 3) \% 26 = 7 = h$$

$$D(h) = (h - 3) \% 26 = (7 - 3) \% 26 = 4 = e$$

$$D(o) = (o - 3) \% 26 = (14 - 3) \% 26 = 11 = l$$

$$D(o) = (o - 3) \% 26 = (14 - 3) \% 26 = 11 = l$$

$$D(r) = (r - 3) \% 26 = (17 - 3) \% 26 = 14 = o$$

∴ P.T. = hello

Example:

Our plain text is: WE WILL MEET IN THE MORNING

Encryption key:3

Plain text	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
Position	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Add 3	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
Cipher text	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T

Plain text	R	S	T	U	V	W	X	Y	Z
Position	17	18	19	20	21	22	23	24	25
Add 3	20	21	22	23	24	25	0	1	2
Cipher text	U	V	W	X	Y	Z	A	B	C

Our cipher text: ZH ZLOO PHHW LO WKH PRUQLQ

❖ Playfair Cipher

- Rules to make playfair matrix of KEY.
1. Arrange in 5*5 matrix.
 2. Double characters are not repeated.
 3. Write down other letters of the character set.
 4. I/J should be on the same position.

Example:**Playfair matrix (key matrix):**

Key=MONARCHY

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

- **Steps of algorithm:**

1. Make a playfair matrix.
2. Split the message (Plain Text) in pair of characters.
 - If a pair is a repeated letter, insert filler like 'X'

Example: TEXT → TE XT

HELLO → HE LL O → HE LX LO

3. Follow the rules for encryption as given below.

- **Rules:**

1. If **both letters fall in the same row**, replace each with letter to right. (wrapping back to start from end)

For example, AR is encrypted as RM

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

2. If **both letters fall in the same column**, replace each with the letter below it (wrapping to top from bottom)

For example, MU is encrypted as CM

3. **Otherwise** each letter is replaced by the letter in the same row and in the column of the other letter of the pair

For Example, SH becomes PB and EA becomes IM

Example:

Key: KEYWORD

Our message to encrypt is : SECRET MESSAGE

Step 1: First of all divide a message in Diagraph (Pair of Character).ADD X to separate duplicate characters.

SE CR ET ME SX SA GE

Step 2: Create Playfair table. First write down Key and then write other Alphabets.

K	E	Y	W	O
R	D	A	B	C
F	G	H	I/J	L
M	N	P	Q	S
T	U	V	X	Z

Step 3: By following above rule(Rule no.3).code for SE is No. Same way byfollowing rule-1,we can get code for CR is RD

So our message is encrypted as : NORDKUNKQZPCND

❖ Hill Cipher

Hill cipher is a polygraphic substitution cipher based on linear algebra.

Encryption

- To encrypt a message using the Hill Cipher, first turn keyword into a key matrix (either in 2x2 matrix or 3x3 matrix).
- Plaintext is split into a column vector.
- Perform matrix multiplication of key matrix with column vector. Do modulo 26 of result.

• 2x2 Example
 Plaintext: "Book"
 Key: $\begin{vmatrix} 3 & 2 \\ 5 & 7 \end{vmatrix}$

Ans → here key matrix $\begin{vmatrix} 3 & 2 \\ 5 & 7 \end{vmatrix}$ determinant $\neq 0$
 $(3 \times 7 - 2 \times 5) = 11 \neq 0$

→ Plain Text is split into Column vector:
 $\begin{vmatrix} B \\ O \end{vmatrix}$ and $\begin{vmatrix} K \\ K \end{vmatrix}$

→ now convert to numeric column vector.
 $\therefore \begin{vmatrix} B \\ O \end{vmatrix} = \begin{vmatrix} 1 \\ 14 \end{vmatrix}$ and $\begin{vmatrix} K \\ K \end{vmatrix} = \begin{vmatrix} 14 \\ 10 \end{vmatrix}$

for "BO", Do matrix multiplication and do modulo 26 of res.
 $\begin{vmatrix} 3 & 2 \\ 5 & 7 \end{vmatrix} \cdot \begin{vmatrix} 1 \\ 14 \end{vmatrix}$
 $= \begin{vmatrix} 3 \times 1 + 2 \times 14 \\ 5 \times 1 + 7 \times 14 \end{vmatrix} \text{ mod } 26$
 $= \begin{vmatrix} 3 + 28 \\ 5 + 98 \end{vmatrix} \text{ mod } 26$
 $= \begin{vmatrix} 31 \\ 103 \end{vmatrix} \text{ mod } 26$
 $= \begin{vmatrix} 5 \\ 25 \end{vmatrix} = \begin{vmatrix} F \\ Z \end{vmatrix}$
 $\therefore BO \rightarrow FZ$

for "OK"
 $\begin{vmatrix} 3 & 2 \\ 5 & 7 \end{vmatrix} \cdot \begin{vmatrix} 14 \\ 10 \end{vmatrix}$
 $= \begin{vmatrix} 3 \times 14 + 2 \times 10 \\ 5 \times 14 + 7 \times 10 \end{vmatrix} \text{ mod } 26$
 $= \begin{vmatrix} 62 \\ 140 \end{vmatrix} \text{ mod } 26$
 $= \begin{vmatrix} 10 \\ 10 \end{vmatrix} = \begin{vmatrix} K \\ K \end{vmatrix}$
 $\therefore OK \rightarrow KK$

$\therefore$ Cipher text is: "FZkk"

❖ **VIGNER CIPHER/ polyalphabetic cipher**

- It is a **polyalphabetic** cipher.
- For Vigenere Cipher, use Vigenere Table.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

• **ENCRYPTION:**

1. To encrypt a message using the Vigenere Cipher you first need to choose a key.
2. Repeat your key until the length of plaintext. Find corresponding row and column from vigenere table.
3. Where these two lines cross in the table, is the ciphertext letter you use.

• **EXAMPLE:**

Plaintext= secret

keyword =bcd

ENCRYPTION:

- First we must generate the keystream, by repeating the letters of the key until it is the same length as the plaintext.

PT	S	E	C	R	E	T
KEY	<u>B</u>	<u>C</u>	<u>D</u>	B	C	D
CT	T	G	F	S	G	W

DECRYPTION:

- Now for decryption, you find T in row of B(Key).
- Means find CipherText letter in the row of key. And then consider corresponding letter of column header for getting plain text back.
- Find T in row of B(Key).the answer is S.(first letter of P.T.)

CT	T	G	F	S	G	W
KEY	<u>B</u>	<u>C</u>	<u>D</u>	B	C	D
PT	S	E	C	R	E	T

EXAMPLE-2:

PT	V	I	G	E	N	E	R	E	C	I	P	H	E	R
KEY	C	O	U	N	T	O	N	C	O	U	N	T	O	N
CT	X	W	A	R	G	S	E	G	Q	C	C	A	S	E

❖ One Time Pad (OTP) (VERNAM (VERMIN/VERMAN) CIPHER)

- It is unbreakable , solid encryption technique.
- Vernam cipher is a stream cipher where the plain text is added with a random stream of data of the same length to generate the encrypted data.

Key characteristics:

- Key must truly random.
- key must be as long as the plaintext, and not repeating
- Key must be used once.
- There should be two copies of the key: One for sender and other for receiver.
- It is also known as One Time Pad (OTP).

Example:

To encrypt the simple message "hello" using the random keystream *jqxyt*.

Cipher Text will be "QUIJH"

Plaintext	h	e	l	l	o
Keystream	J	Q	X	Y	T
Ciphertext	Q	U	I	J	H

- **Example -2:**
- **Plaintext :** HOW ARE YOU
- **Onetime pad(Key) :** NCB TZQ ARX

1. Plain text	H	O	W	A	R	E	Y	O	U
	7	14	22	0	17	4	24	14	20
	+								
2. One-time pad	13	2	1	19	25	16	0	17	23
	N	C	B	T	Z	Q	A	R	X
3. Initial Total	20	16	23	19	42	20	24	31	43
4. Subtract 26, if > 25	20	16	23	19	16	20	24	5	17
5. Ciphertext	U	Q	X	T	Q	U	Y	F	R

[For Decryption: Do (C.T. - Key) Modulo26 for getting P.T.]

❖ Rail fence cipher

(TRANSPPOSITION TECHNIQUES)

Here, Plaintext symbols are rearranged to produce Ciphertext.(**reordering of letters of PlainText** to calculate Cipher text –in transposition techniques)

❖ **Example:** Rail fence cipher**Rail fence cipher(Explain with example)**

- Rail fence cipher is a type of transposition cipher.
- It involves rearranging of letters in plain text to encrypt the message.

Example:

Suppose we want to encrypt the message “**INFORMATION SECURITY**” using a rail fence cipher with encryption **key 3**. Here is how we would proceed.

Plain Text: - INFORMATION SECURITY

Number of Rails/Key: - 3

- i. Arrange the plain text characters in an array with 3 rows(the key determines the number of rows), forming a zig-zag pattern:

I	N	F	O	R	M	A	T	I	O	N	S	E	C	U	R	I	T	Y
I				R				I				E				I		
	N		O		M		T		O		S		C		R		T	
		F			A				N					U				Y

- ii. Then concatenate the non-empty characters from the rows to obtain the

ciphertext: IRIEINOMTOSCRTFANUY

For practice, try encrypting the plaintext “a new semester begins with football and moving vans “using a rail fence cipher with key 5 and see if you obtain the ciphertext

❖ **Decryption:**

Determine the Zigzag Pattern: - The number of columns in the rail fence cipher remains equal to the length of plain-text message. And the key corresponds to the number of rails.

Hence, the rail matrix can be constructed accordingly. Once we've got the matrix we can figure-out the spots where texts should be placed.

Fill in the Encrypted Message: - Fill the cipher-text row wise.

Read the Decrypted Message: -Read the characters diagonally from the top-left corner to the bottom-right corner, reversing direction when needed. The resulting sequence is the decrypted message.

Example:

Ciphertext: - IRIEINOMTOSCRTFANUY

Number of Rails/Key: - 3

*				*				*				*				*		
	*		*		*		*		*		*		*		*		*	
		*				*				*			*					*
I				R				I				E				I		
	N		O		M		T		O		S		C		R		T	
		F				A				N				U				Y

Decrypted Message: - INFORMATION SECURITY

2.3 Steganography

❖ STEGNOGRAPHY

- **Steganography means hiding a message within another message or image.**
- Steganography is the art and science of hiding secret data within an ordinary, non-secret file or medium to conceal the fact that a communication is even taking place.
- It differs from cryptography because it hides the *existence* of the message, rather than making the message unreadable.
- Steganography is the art of **hiding a message**, image, or file **within another message**, image, or file.
- For example, the hidden message may be in invisible ink between the visible lines of a private letter.
- **Examples of Steganography:**

1)Character Marking

2)Invisible Ink

3)Pin punctures

➤ **Methods of Steganography**

- **Digital Steganography:**

We can insert data or we can hide data in the image by replacing bits of image.

Most common technique are:

1)LSB,

2)DCT and

3)Append type.

- **LSB: Least Significant Bit**

-Replace LSB bit with a bit from hidden data.

-LSB has smallest effect on the amount of color.

-Replacing LSB to hidden data will have small effect on the picture.

- **DCT stands for Discrete Cosine Transform**

-It works by calculating the frequencies of the image then replace some of them.

- **Append:**

-Instead of hide the data in the photo by manipulating the picture, Append algorithm appends the data to the end of the file as appending.

-This algorithm will change the size of the file.

➤ **Types of Steganography**

Here is a comprehensive breakdown of the **types of steganography based on the carrier medium** used.

1. Image Steganography

- This is the most popular form of steganography due to the massive amount of digital image sharing online. Data is embedded by subtly altering the digital properties of the image.
- **Least Significant Bit (LSB) Insertion:** Alters the last bit of a pixel's color byte (e.g., changing a color value from 255 to 254). The human eye cannot detect this minuscule color shift.
- **Discrete Cosine Transform (DCT):** Alters the mathematical coefficients used in JPEG compression. This embeds data in the frequency domain, making it highly resilient to image resizing or cropping.
- **Palette Modification:** Hides data by altering the color palette list inside indexed images like GIFs.

2. Audio Steganography

- Secret data is embedded into digital audio signals. It is highly complex because the human auditory system is incredibly sensitive to sound variations.
- **Low-Bit Encoding:** Similar to LSB in images, this modifies the least significant bits of the audio amplitude files.
- **Phase Coding:** Replaces the phase of an initial audio segment with a reference phase that represents the secret data.
- **Echo Hiding:** Injects a subtle, unnoticeable echo into the audio track. The delay between the original sound and the echo represents the hidden binary data.

3. Video Steganography

Video steganography treats a video as a continuous stream of images (frames) and audio. It allows for the concealment of much larger data payloads compared to other media.

- **Frame-by-Frame Embedding:** Treats individual video frames as static images and applies standard image steganography (like LSB) to them.
- **Motion Vector Embedding:** Hides data inside the motion vectors used during video compression (e.g., MPEG, H.264) to predict pixel movement between frames.

4. Text Steganography

- Text steganography involves altering the formatting, arrangement, or structure of words within a document. It requires the least amount of memory but offers a very small hiding capacity.
- **White Space Manipulation:** Injects hidden binary code using unique combinations of single spaces, double spaces, and tabs at the ends of lines.
- **Line/Word Shifting:** Slightly shifts the vertical lines or horizontal spacing of specific words in a PDF or printed text.

- **Syntactic/Semantic Methods:** Uses specific grammar structures or replaces words with synonyms at predetermined intervals to encode a message.

5. Network (Protocol) Steganography

- This type hides data inside the actual network packets passing through a system, making it nearly impossible to detect with standard file scans.
- **Header Manipulation:** Injects secret data bits into unused, optional, or flexible fields within network headers (such as TCP, UDP, or IP headers).
- **Packet Timing:** Alters the delay or arrival intervals between consecutive network packets to transmit binary code (e.g., a short delay represents 0, a long delay represents 1).

6. Physical & Digital Printed Steganography

- This bridges the gap between digital files and physical objects.
- **Microdots:** Shrinks a text or image down to the size of a typewriter period, making it viewable only under a microscope.
- **Digital Watermarking / Printer Dots:** Fine, yellow tracking dots printed invisibly by commercial laser printers onto physical paper to encode serial numbers and timestamps.

2.4 Symmetric Cryptography: Data Encryption Standard- Structure, Advantages and Disadvantages

❖ Data Encryption Standard (DES):

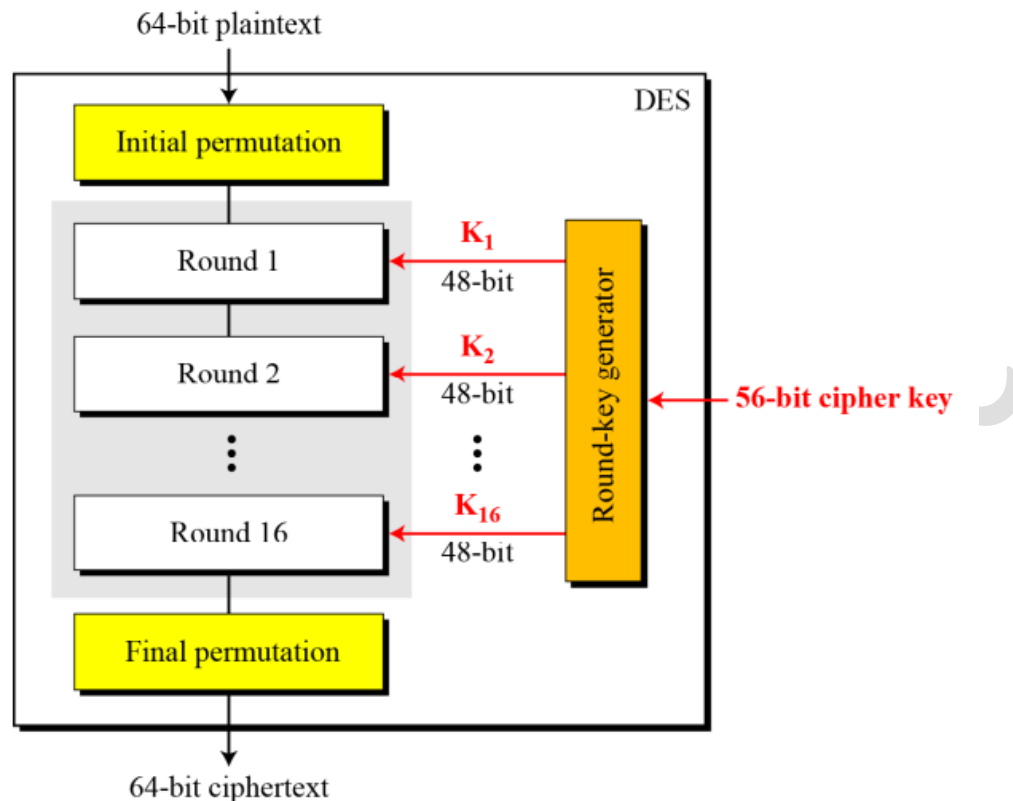
The Data Encryption Standard (DES) is a **symmetric-key block cipher** published by the National Institute of Standards and Technology (NIST).

DES Structure:

The Data Encryption Standard (DES) is a **block cipher**.

The encryption process is made of two permutations (P-boxes), which we call initial and final permutations, and sixteen Feistel rounds.

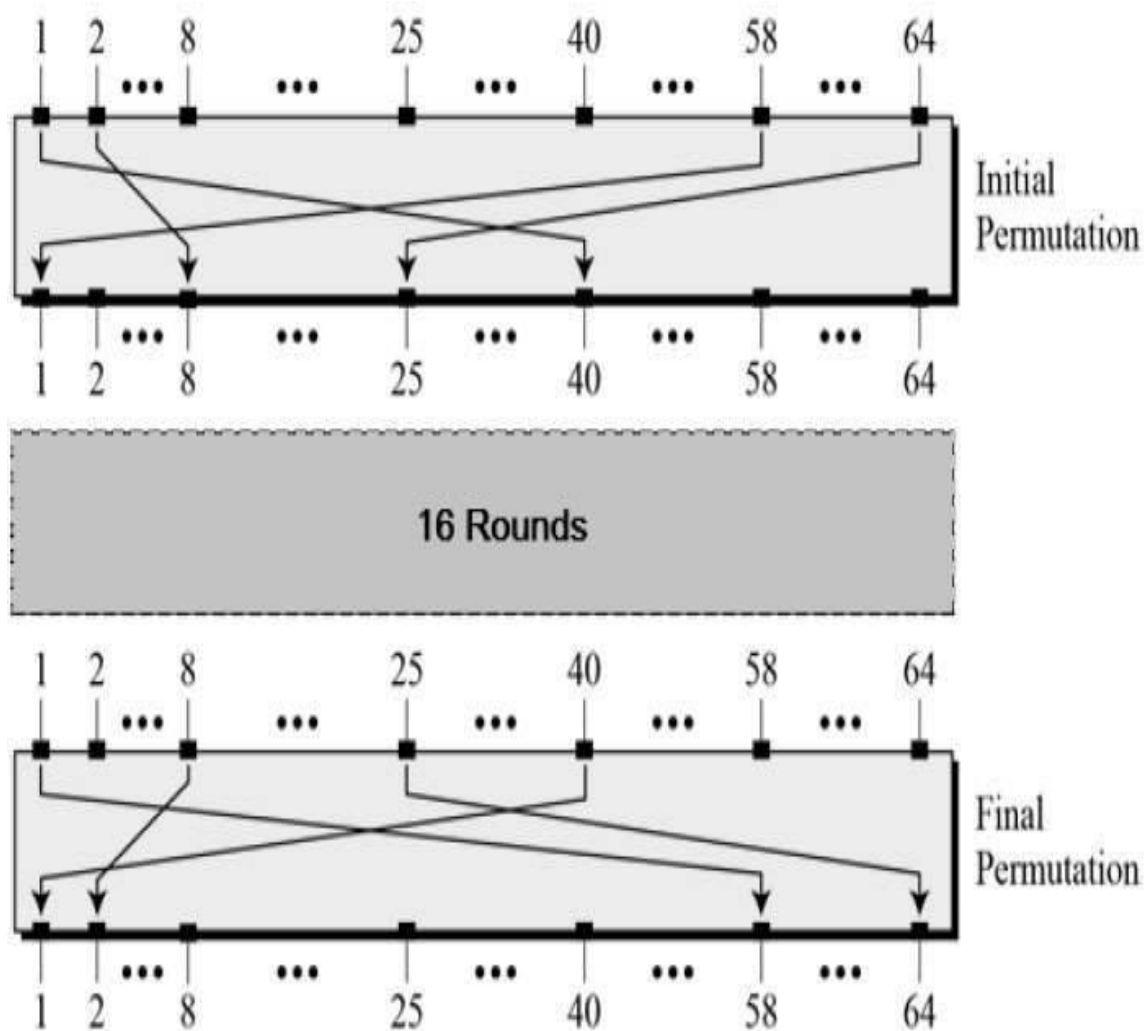
General Structure of DES is shown in figure:



- ✓ DES is an implementation of a Feistel Cipher.
- ✓ It uses 16 round Feistel structure. The block size is 64-bit.
- ✓ Though, key length is 64-bit, DES has an effective key length of 56 bits, since 8 of the 64 bits of the key are not used by the encryption algorithm (function as check bits only).
- ✓ Since DES is based on the Feistel Cipher, all that is required to specify DES is –
 - Round function
 - Key schedule
 - Any additional processing – Initial and final permutation

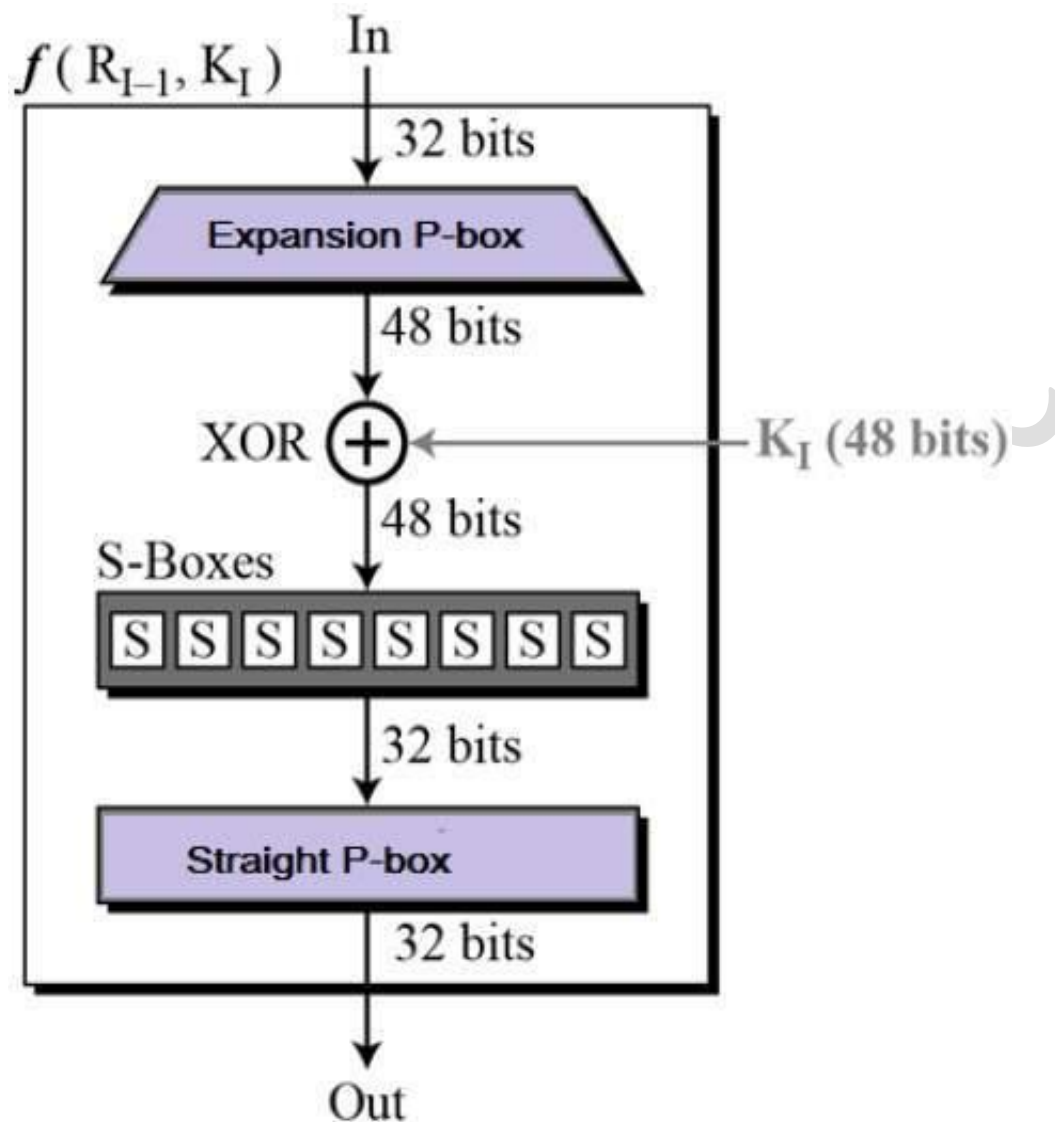
Initial and Final Permutation

- The initial and final permutations are straight Permutation boxes (P-boxes) that are inverses of each other. They have no cryptography significance in DES.
- The initial and final permutations are shown as follows –



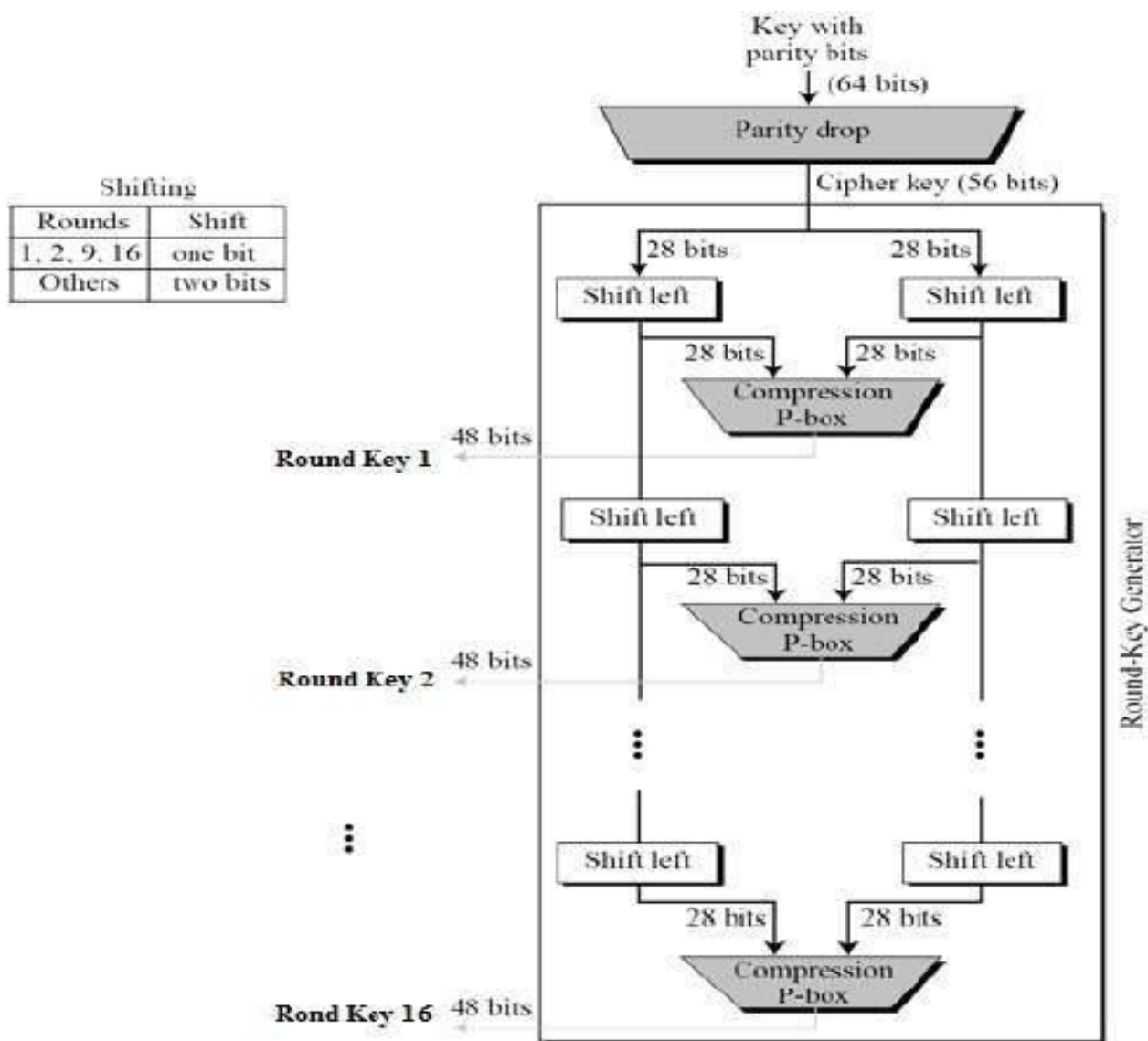
Round Function

The DES function f applies a 48-bit key to the rightmost 32 bits to produce a 32-bit output.



Key Generation

The round-key generator creates sixteen 48-bit keys out of a 56-bit cipher key.



Advantage of DES

- DES uses the symmetric-key algorithm, thus, it is possible to perform encryption and decryption by a single key with the same algorithm.
- It is more efficient in hardware, showing a higher and faster implementation.
- DES is relatively fast and efficient, making it suitable for use in a wide range of.
- DES has a relatively small key size, which makes it easier to use and store.
- It's not a group cipher, hence DES instances can be applied many times to a plaintext.(2DES 3DES).

- The DES satisfies both the desired properties of block cipher. These two properties make cipher very strong.

Avalanche effect – A small change in plaintext results in the very great change in the ciphertext.

Completeness – Each bit of ciphertext depends on many bits of plaintext.

Disadvantage of DES

- Because DES uses a smaller key, it is less secure.
- The DES algorithm is less efficient when implemented in software, resulting in slower performance.
- DES offers a lower level of security due to its 56-bit key, which can be feasibly broken by a brute-force attack.
- Now in the age of parallel computing, breaking DES has become easy with the help of brute force attack which was impossible during that time.(it's possible to brute-force in finite time on modern processors).
- In a new technology, it is improving a several possibility to divide the encrypted code, therefore AES is preferred than DES.